Muhammad Bilal Ali

Pittsburgh, PA | Linkedin | Portfolio | Github

Education

Carnegie Mellon University — Pittsburgh, US
Masters in Privacy Engineering and AI Governance — *Aug 2025 to Dec 2026*

- Coursework: Engineering Privacy in Software, Usable Privacy and Security, Distributed Systems (Java, Apache Spark), Deep Learning, Database Management

Lahore University of Management Sciences — Lahore, Pakistan
Bachelor of Science in Computer Science; Minor in Economics — *Aug 2020 to Jul 2025*

- Coursework: Algorithms, Data Structures, Network Security, Computer Organization, Artificial Intelligence, Cloud Development
- **President, Publications Society:** Led a 20+ member editorial team, set content strategy for publications reaching the whole undergraduate body, and secured brand sponsorships for society events.
- **Teaching Assistant:** Reviewed and debugged code from **500+ students** across five courses over four semesters.

Experience

Carnegie Mellon University, S3D Department — Pittsburgh, US
Research Assistant | LLM Pipelines, Privacy Threat Modeling, Python — *Jun 2026 to Aug 2026*

- Built an **LLM pipeline that finds privacy threats** in system designs and source code using **LINDDUN Pro**, linking each to the GDPR, CCPA, or COPPA requirement it affects. **Presented at IEEE ISoPE 2026**.
- Evaluated it against **99 expert written threats** from **NIST SP 1800** on **recall** (missed threats), **precision** (false alarms), and **citation correctness**; used **ablation studies** to trace errors to the model, retrieved context, or prompts.

INDITECH MIDDLE EAST — Dubai, UAE
Data and Analytics Intern | SQL, Python, AWS, Power BI, Tableau — *Jun 2023 to Aug 2023*

- Wrote **100+ SQL queries** and built **AWS data pipelines** merging separate sales and inventory systems; traced the root causes of unreliable reports and fixed them, applying access controls and data minimization.
- Built **Power BI and Tableau dashboards** for management that led to **fewer overstock incidents**.

Projects and Research

EasyDine: Recipe Sharing App | easydine.org — *In review: Google Play and Apple App Store*

- Designed and built a **full mobile app end to end**: UI design, user accounts and authentication, recipe sharing, backend services, database design, and the website; now **submitted to both app stores** for review.
- Added AI features (a chatbot and **video reels converted into recipes**) and a **live ingredient price pipeline** pulling from external sources; prepared the **privacy disclosures** both stores require.

Privacy Preference Tool for Smart Home Devices — *CMU Capstone with CableLabs, Sep to Dec 2026*

- Co-developing a prototype of the proposed **IETF Privacy Preference Declaration** framework so households can control what smart devices do with their data: a web tool, a preference vocabulary, and a **JSON Schema validator**.

Membership Inference Attack on a Fine-Tuned LLM — *PyTorch, Hugging Face, PEFT*

- Attacked a **GPT-2 model fine-tuned with LoRA** to reveal its training data; trained **4 shadow models** matched to the target, scored by **per-token loss and perplexity**, and evaluated with **ROC AUC and TPR at 1% FPR**.

Second Brain: RAG and AI Agent Tooling — *Python, FastAPI, Qdrant, MCP*

- Built a **RAG service** that searches PDFs, Notion, YouTube, and web pages in one query, combining **Qdrant vector search, BM25, and rank fusion** with **cross-encoder re-ranking** for more relevant results.
- Connected it to Claude through a custom **MCP server** with multi-query expansion, serving streaming answers via FastAPI that **cite the source** behind every claim.

Skills

Languages and Frameworks: Python, Java, C/C++, Go, SQL, JavaScript, React, Node.js, FastAPI, PyTorch, Hugging Face
Privacy and Security: Threat modeling (LINDDUN Pro), privacy reviews, privacy attacks, access controls, GDPR, CCPA, COPPA
AI and Data: LLM pipelines, AI agents (MCP), RAG, context engineering, LLM evaluation, Apache Spark, Power BI, Tableau
Infrastructure and Tools: PostgreSQL, MongoDB, Qdrant, Docker, AWS, Git, Linux